The Saudi Legal Corpus for AI: An Auditable, Official-Source-Grounded, Article-Level Corpus of Saudi Arabian Legislation

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Abstract

Legal natural language processing has advanced rapidly for English and, more recently, for European languages, but Arabic — and Saudi Arabian law in particular — remains critically under-resourced: no openly documented, machine-readable, article-level corpus of Saudi legislation has previously been described in the literature. We present the **Saudi Legal Corpus for AI**, a corpus of **291 legislative tracks** (201 statutory laws, 64 statutory regulations, 24 implementing regulations, and associated procedural rules) covering the Kingdom of Saudi Arabia's principal legislation, structured into **15,689 article-level records** ($\sim$1.2 million Arabic tokens) in a unified, LLM-ready retrieval index. The corpus is built on three design commitments that distinguish it from web-scraped legal collections: (1) *official-source grounding* — every track records its issuing authority, official portal or gazette source, and per-record provenance labels (251 distinct provenance statuses), organized into a four-tier source-verification taxonomy; (2) *auditability* — 376 read-only, idempotent validators enforce schema, counts, and hashes, and all derived layers are deterministic and re-runnable; (3) *the governing-text principle* — the official Arabic text governs, and all other language layers are explicitly non-authoritative references. Beyond the text itself, the corpus ships machine-readable analytical layers that are, to our knowledge, the first of their kind for Saudi law: a cross-reference graph (3,607 extracted citations, 585 of them inter-law), a hand-classified supersession graph (102 edges), a glossary of 1,920 statutorily defined terms with 3,318 definitions, an embedding-ready chunking layer (16,304 chunks), and a manually verified retrieval evaluation set of 519 gold queries, on which a deterministic lexical baseline achieves 93.3% top-1 accuracy (MRR@5 = 0.956), leaving informative headroom for neural Arabic legal retrieval. We describe the corpus construction methodology, report detailed statistics, and discuss intended uses, limitations, and the legal and ethical boundaries of releasing structured national legislation for AI research.

1 Introduction

Large language models are increasingly consulted for legal questions, and retrieval-augmented generation (RAG) over authoritative legal text is the standard mitigation for hallucinated law. That mitigation, however, is only available where authoritative legal text exists in machine-readable, retrieval-ready form. For the law of the Kingdom of Saudi Arabia — a G20 economy undergoing the most significant codification wave in its history, including a new Civil Transactions Law (721 articles), a new Companies Law (281 articles), a new Evidence Law, and a new Personal Status Law — no such resource has been publicly documented. Saudi legislation is published officially in the *Umm Al-Qura* gazette and on government portals as human-readable pages and PDFs; it is not distributed as structured, article-level, provenance-annotated data.

This paper presents the **Saudi Legal Corpus for AI**, which closes that gap. The corpus structures the principal body of Saudi legislation into canonical article-level JSON with explicit provenance, wraps it in deterministic validation, and derives from it a unified retrieval index, a chunking layer, citation and supersession graphs, a glossary of statutory definitions, and a manually verified retrieval benchmark.

Our contributions are:

1. **The corpus itself**: 291 legislative tracks, 15,689 article-level records, $\sim$1.2M Arabic tokens, covering statutory law, regulations, implementing regulations, procedural rules,

model contracts, and official annexes across commercial, civil, criminal, labor, financial, real-estate, technology, health, and administrative domains (Section 5).

2. **A methodology for auditable legal corpora**: official-source grounding with per-record provenance labels, a four-tier source-verification taxonomy, a freshness manifest that documents known staleness risks instead of hiding them, and 376 read-only idempotent validators (Section 4). We argue this methodology is a reusable template for legal corpora in any jurisdiction without a machine-readable official gazette.

3. **Derived analytical layers** unavailable anywhere else for Saudi law: a cross-reference graph, a hand-classified supersession graph, and a statutory-definition glossary (Section 6).

4. **A retrieval evaluation resource**: 519 manually confirmed gold queries over the unified index, with a deterministic lexical baseline (top-1 = 93.3%, MRR@5 = 0.956) that future neural Arabic legal retrievers can be measured against (Section 7).

Throughout, the corpus maintains a strict legal-status boundary: it is **not** an official government publication, contains **no official translation**, and the **official Arabic text governs**; English and Chinese layers are non-authoritative references. We return to these boundaries in Section 10.

2 Related Work

Legal corpora and benchmarks. Large legal text collections and benchmarks now exist for English and several European jurisdictions: the Pile of Law (Henderson et al., 2022), MultiLegalPile (Niklaus et al., 2024), LexGLUE (Chalkidis et al., 2022), LEXTREME (Niklaus et al., 2023), CUAD (Hendrycks et al., 2021), and LegalBench (Guha et al., 2023). These resources are dominated by English and European languages; none includes Saudi legislation, and Arabic is at most marginally represented.

Arabic legal NLP. Arabic NLP infrastructure has matured (e.g., Antoun et al., 2020, Obeid et al., 2020), and domain-specific models such as AraLegal-BERT (Al-Qurishi et al., 2022) have been trained on proprietary Arabic legal text. However, openly documented *data resources* for Arabic law — in particular structured, provenance-annotated statute corpora — remain scarce. To our knowledge, no prior published resource provides article-level, machine-readable coverage of Saudi legislation with documented official-source provenance.

Legal document standards and citation networks. Standards such as Akoma Ntoso (Palmirani and Vitali, 2011) and the European Legislation Identifier define structural markup for legislation; our canonical JSON is deliberately simpler (flat article-level records with rich retrieval metadata) but is convertible to such standards. A separate literature applies network science to legal citation structure (Coupette et al., 2021, Fowler et al., 2007, Katz and Bommarito, 2014); the cross-reference and supersession graphs we release (Section 6) are, to our knowledge, the first machine-readable inputs enabling that line of work for Saudi — and more broadly Gulf — legislation.

3 Source Material: The Saudi Legislative Landscape

Saudi legislation is enacted primarily by royal decree (*niẓām* — conventionally "law" or "system") and elaborated by implementing regulations issued by ministers or authorities, alongside procedural rules, ministerial decisions, and official model forms. The authoritative publication channel is the official gazette *Umm Al-Qura*; consolidated texts are additionally published on official portals, principally the Bureau of Experts at the Council of Ministers (`laws.boe.gov.sa`) and the Ministry of Justice legal portal (`laws.moj.gov.sa`), as well as issuing ministries' and authorities' own sites.

Each corpus **track** corresponds to one legislative instrument (one law, one regulation, one set of rules) and records its Arabic and English display names, issuing authority, official source URL(s), publication dates (Hijri and Gregorian where available), language layers, record counts, and validator targets, in a central machine-readable registry (registry version 3.0).

4 Corpus Design and Methodology

4.1 Design principles

The corpus is built around four commitments:

1. **Official Arabic text governs.** Every record carries the official Arabic text verbatim (`text_ar`); no summarization, normalization, or editorial rewriting is applied to the governing text. All non-Arabic layers are explicitly flagged as non-authoritative.

2. **Provenance is per-record and honest.** Every record in the unified index carries a `text_status` provenance label describing exactly how its text was obtained and verified (e.g., cross-checks between the Ministry of Justice portal API and published official PDFs). There are 251 distinct provenance labels — deliberately fine-grained rather than collapsed, because collapsing provenance destroys auditability. Where the primary portal was found stale at build time (e.g., an unincorporated amendment), that is documented, not silently patched: the freshness manifest flags 16 of 291 tracks with a known source-staleness risk.

3. **Validation is read-only, deterministic, and idempotent.** The repository ships 376 validator targets (of 476 total build targets) that check schema conformance, record counts, hashes, layer boundaries, and cross-layer consistency without mutating any data. Derived-layer generators make no network calls and use no wall-clock time, so every layer is exactly reproducible from the committed sources.

4. **Derived layers are additive projections.** The unified index, chunking layer, graphs, glossary, tiers, and freshness manifest are all read-only projections over the canonical track files. None of them alters legal text; each records what it was generated from.

4.2 Source verification tiers

Because Saudi Arabia has no machine-readable official gazette, source quality varies by track and must be represented rather than assumed. Every track is assigned to one of four verification tiers (Table 1). This taxonomy lets downstream users filter by evidential strength — a RAG deployment may restrict itself to Tiers 1–2, while a coverage-oriented study may accept all four with the caveats attached.

Tier	**Tracks**
1 — primary, multi-source	127
2 — primary + secondary cross-verified	88
3 — secondary multi-source only	39
4 — single-source or mixed confidence	37

Table 1: Source-verification tier assignments (291 tracks).

Corpus family	**Tracks**
Statutory laws	201
Statutory regulations	64
Implementing regulations	24
Procedural rules	1
Closure-audit track	1
Total	**291**

Table 2: Track-level composition of the corpus.

4.3 Record schema

The atomic unit is the **article-level record**. In the unified retrieval index each record carries: a stable `record_id`; corpus/track identifiers (`corpus`, `law_id`, `law_component`); the official Arabic law title and article number; generated retrieval metadata (`llm_title_ar`, `retrieval_title_ar`, `keywords_ar`, `search_queries_ar`); the verbatim governing text (`text_ar`); the provenance label (`text_status`); and the source layer file it was projected from. Records are self-contained: a single retrieved record is sufficient context to cite the law, component, and article.

5 Corpus Contents and Statistics

All figures are computed by the accompanying read-only script and stored alongside the corpus.

5.1 Track-level composition

Track composition is shown in Table 2. The registry counts 16,753 records in total, of which 15,858 are primary Arabic governing records, 614 are reference records, and 281 are internal reference records (the Chinese internal layer for the Companies Law).

Measure	Value
Article-level records	15,689
Distinct legislative instruments	283
Distinct corpora	199
Arabic tokens (whitespace)	1,206,097
Characters	7,123,487
Distinct provenance labels	251

Table 3: The unified retrieval index at a glance.

Component	Records
Law articles	8,384
Regulation articles	4,655
Implementing-regulation articles	2,113
Procedural manuals	135
Model contract forms	102
Model work regulation	75
Recruitment services rules	72
Other rules, guides, annexes	153

Table 4: Unified-index records by component type.

5.2 The unified retrieval index

The unified LLM retrieval index projects the retrieval-relevant records into one file; Table 3 summarizes it and Table 4 breaks records down by component type.

Coverage spans the core commercial stack (Companies Law, Bankruptcy, Commercial Courts, Capital Market, Banking Control), the new codification wave (Civil Transactions — 721 articles, Evidence, Personal Status), criminal and procedural law (Criminal Procedure, Sharia Procedure and its 637-article regulation, Enforcement), labor and social insurance (including official annexes: violation tables, model contracts), technology and data (the Personal Data Protection Law and its regulations, E-Commerce, Electronic Transactions, Cybersecurity), tax and zakat (VAT, RETT, Income Tax, Zakat), real estate (registration, brokerage, finance, off-plan sales, contributions), health, environment, media, transport, and the institutional statutes of major authorities.

5.3 Multilingual layers

The Saudi Companies Law (Royal Decree M/132, 1443H; 281 articles) is the first track with full multilingual treatment: the governing Arabic layer, a full English LLM layer (281 records), an English reference layer (281 records), and an internal Chinese reference layer that passed a documented four-phase remediation and QA program, closed by a repository-level audit. The corpus explicitly does **not** claim trilingual alignment or an official translation; the multilingual layers exist to support cross-lingual access research under the governing-text principle.

6 Derived Analytical Layers

6.1 Cross-reference graph

A deterministic extractor scans all 15,689 records for statutory citations, yielding **3,607 references**: 3,022 intra-law and **585 inter-law** citations, emitted with the raw citation text, source/target identifiers, and a confidence label (3,087 high, 520 medium; 0 ambiguous-scope). The 585 inter-law edges span 191 distinct source tracks and 316 distinct cited target names; the most-cited targets are the Labor Law (34 inbound citations), the Companies Law (30), the Bankruptcy Law (19), and the Sharia Procedure Law (14). This is, to our knowledge, the first machine-readable citation network over Saudi legislation, and it directly enables network-science studies of the Saudi legal system (centrality, clustering, dangling references to superseded instruments).

6.2 Supersession graph

A separate layer records repeal/replacement relationships between instruments: **102 edges**, each classified *by hand* against the corpus's own documented text (registry notes and official-source metadata) — never inferred from decree age or topical similarity. Ambiguous cases and title-collision non-repeals are kept in their own sections rather than forced into edges. Combined with the citation graph, this supports temporal legal dynamics research (e.g., which live laws still cite repealed ones).

6.3 Statutory-definition glossary

Saudi laws conventionally open with a definitions article. A parser over these articles yields a corpus-wide glossary of **1,920 terms** with **3,318 definitions**, extracted from the 214 tracks whose definitions articles parse cleanly. Because the same term (e.g., “establishment”, “competent authority”) is defined independently by many laws, the glossary is the raw material for studying definitional fragmentation and terminological consistency across a national legal system.

6.4 Freshness manifest

A deterministic, network-free manifest surveys every track's recorded source URLs, fetch dates, and known discrepancies, and flags **16 tracks** with a documented risk that the primary portal's consolidated text was stale at build time. A separate live checker (explicitly outside the QA gate) can re-probe reachability and drift.

7 Retrieval Index, Chunking, and Baseline Evaluation

7.1 Chunking layer

For embedding pipelines, a chunking layer segments the unified index into **16,304 chunks** sized in whitespace words (deliberately not tied to any specific embedding tokenizer). 15,344 records fit in a single chunk; 345 long articles are split, with each chunk carrying its source record identifier, article number, chunk index, and exact character offsets into the source text, so any vector hit is traceable to its governing article. No embeddings are computed in-repository; the layer is pure, traceable segmentation.

7.2 Gold-query evaluation set

The retrieval evaluation pack contains **519 Arabic queries**, each with a manually confirmed gold article (corpus, component, article number). Golds were confirmed against the articles' own text or official structure — not reverse-engineered from search output — so the metrics honestly measure the searcher. Queries are categorized: 384 topical, 61 provision-lookup, 55 definitional, and 19 in smaller categories (governance, rights, procedural, and deliberately hard cases).

7.3 Lexical baseline results

A deterministic lexical scorer over each record's retrieval metadata (weighted keyword/title/query matching; no embeddings, no learned parameters) achieves the results in Table 5.

Metric	**Value**
Top-1 accuracy	93.3% (484/519)
Top-3 accuracy	97.5% (506/519)
Top-5 accuracy	98.8% (513/519)
MRR@5	0.956

Table 5: Deterministic lexical baseline on the 519-query evaluation set.

Six queries miss the top-5 entirely; these are published per-query, making error analysis reproducible. The strength of this simple baseline reflects the retrieval metadata engineered into each record; the remaining headroom — and, more importantly, robustness to paraphrase and dialectal variation not present in the gold queries — is exactly the space in which neural Arabic legal retrievers should be evaluated. We release the queries, golds, per-query results, and the runner script to support that comparison.

8 Intended Uses

- **RAG grounding for Saudi-law assistants**, with per-record provenance and verification tiers enabling deployments to filter by evidential strength.
- **Arabic legal-NLP research**: retrieval, summarization, question answering, terminology extraction, and cross-lingual legal access, over a morphologically rich, formulaic register of Modern Standard Arabic.
- **Computational legal studies**: citation-network analysis, definitional consistency, law-vs-regulation delegation studies, temporal dynamics of the 2022–2023 codification wave.
- **Benchmarking factuality**: the corpus provides ground truth for measuring LLM hallucination on Saudi legal questions.
- **A methodological template** for building auditable legal corpora in jurisdictions without machine-readable official gazettes.

9 Limitations

- **Not exhaustive.** 291 tracks cover the principal legislation, but not every ministerial decision, circular, or municipal instrument;

coverage is documented per-track rather than claimed complete.

- **Consolidation risk.** Official portals themselves sometimes lag amendments; 16 tracks carry a documented staleness-risk flag, and 37 tracks sit in the lowest verification tier. The corpus documents these risks; it cannot eliminate them.
- **Extraction layers are heuristic.** The cross-reference extractor and glossary parser are deterministic but pattern-based; their published caveats (confidence labels, skipped tracks) bound but do not remove extraction error. The supersession graph is hand-classified and therefore conservative and possibly incomplete.
- **The evaluation set is lexically close to the corpus.** The 519 gold queries were written against the corpus's own terminology; they measure in-vocabulary retrieval well, paraphrase robustness less so.
- **Multilingual depth is uneven.** Only the Companies Law has full English and internal Chinese layers; the corpus is Arabic-first by design.

10 Ethical and Legal Considerations

The corpus contains enacted national legislation — public legal facts, not personal data. Nevertheless, three boundaries are enforced repository-wide and repeated here: the corpus is **not an official government publication**; its non-Arabic layers are **not official translations**; and nothing in it is **legal advice** — the only binding text is the Arabic original as published in *Umm Al-Qura*. The structuring apparatus (schemas, scripts, derived layers, metadata) is released under the MIT license; the underlying statutory texts remain public legal enactments of the Kingdom of Saudi Arabia, and the corpus asserts no ownership over them. A misuse consideration specific to legal corpora is over-trust: shipping provenance labels, verification tiers, and staleness flags *inside* the data — rather than in documentation users never read — is a deliberate design choice so that downstream systems can surface uncertainty instead of laundering it.

11 Conclusion and Future Work

We presented the Saudi Legal Corpus for AI: 291 legislative tracks and 15,689 article-level records of Saudi legislation with per-record provenance, tiered source verification, deterministic validation, derived citation/supersession/glossary layers, and a manually verified 519-query retrieval benchmark with a strong lexical baseline. Future work proceeds on three axes: (1) neural baselines — Arabic and multilingual embedding models evaluated on the released benchmark, including paraphrase-robustness extensions of the query set; (2) analytical studies over the released graphs — legislative network structure, definitional fragmentation, and the temporal dynamics of the Saudi codification wave; and (3) coverage growth — onboarding further instruments through the same profile-driven factory, with amendment-tracking as the hardest open problem for any legal corpus in a jurisdiction without a machine-readable gazette.

Data Availability

The corpus, all derived layers, all validators, the evaluation pack, and the statistics script for this paper are available in the repository `al3obdi/saudi-legal-co` *[TODO before submission: mint a versioned, DOI-carrying archival release (e.g., Zenodo) and cite it here.]*

References

Muhammad Al-Qurishi, Sarah Souissi, and Riad Souissi. AraLegal-BERT: A pretrained language model for Arabic legal text. In *Proceedings of the Natural Legal Language Processing Workshop 2022*, Abu Dhabi, United Arab Emirates, 2022. Association for Computational Linguistics.

Wissam Antoun, Fady Baly, and Hazem Hajj. AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT)*, Marseille, France, 2020. European Language Resources Association.

Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Martin Katz, and Nikolaos Aletras. LexGLUE:

A benchmark dataset for legal language understanding in English. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, Dublin, Ireland, 2022. Association for Computational Linguistics.

Corinna Coupette, Janis Beckedorf, Dirk Hartung, Michael Bommarito, and Daniel Martin Katz. Measuring law over time: A network analytical framework with an application to statutes and regulations in the United States and Germany. *Frontiers in Physics*, 9, 2021.

James H. Fowler, Timothy R. Johnson, James F. Spriggs, Sangick Jeon, and Paul J. Wahlbeck. Network analysis and the law: Measuring the legal importance of precedents at the U.S. Supreme Court. *Political Analysis*, 15(3):324–346, 2007.

Neel Guha, Julian Nyarko, Daniel E. Ho, Christopher Ré, et al. LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023.

Peter Henderson, Mark S. Krass, Lucia Zheng, Neel Guha, Christopher D. Manning, Dan Jurafsky, and Daniel E. Ho. Pile of Law: Learning responsible data filtering from the law and a 256GB open-source legal dataset. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022), Datasets and Benchmarks Track*, 2022.

Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. CUAD: An expert-annotated NLP dataset for legal contract review. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021), Datasets and Benchmarks Track*, 2021.

Daniel Martin Katz and Michael J. Bommarito. Measuring the complexity of the law: The United States Code. *Artificial Intelligence and Law*, 22(4):337–374, 2014.

Joel Niklaus, Veton Matoshi, Pooja Rani, Andrea Galassi, Matthias Stürmer, and Ilias Chalkidis. LEXTREME: A multi-lingual and multi-task benchmark for the legal domain. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, Singapore, 2023. Association for Computational Linguistics.

Joel Niklaus, Veton Matoshi, Matthias Stürmer, Ilias Chalkidis, and Daniel E. Ho. MultiLegalPile: A 689GB multilingual legal corpus. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, Bangkok, Thailand, 2024. Association for Computational Linguistics.

Ossama Obeid, Nasser Zalmout, Salam Khalifa, Dima Taji, Mai Oudah, Bashar Alhafni, Go Inoue, Fadhl Eryani, Alexander Erdmann, and Nizar Habash. CAMeL Tools: An open source Python toolkit for Arabic natural language processing. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, Marseille, France, 2020. European Language Resources Association.

Monica Palmirani and Fabio Vitali. Akoma-Ntoso for legal documents. In *Legislative XML for the Semantic Web*, pages 75–100. Springer, 2011.